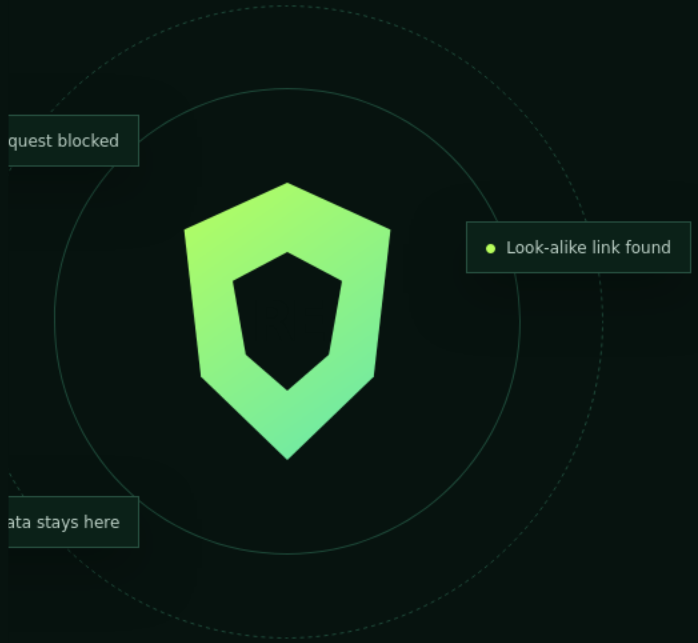
ON-DEVICE SOCIAL-ENGINEERING FIREWALL

Rakshak Edge

A private safety check before the user clicks, pays, or shares a secret



SNAPDRAGON X SERIES

Hariharan C | 2026

The critical moment happens before the tap

A suspicious message exploits urgency and trust. Sending it to a cloud checker can expose the same private content the user is trying to protect.

01

Private content

OTP messages, account alerts, payment handles, and personal context.

02

Fast manipulation

Threats and rewards push the user to act before checking another channel.

03

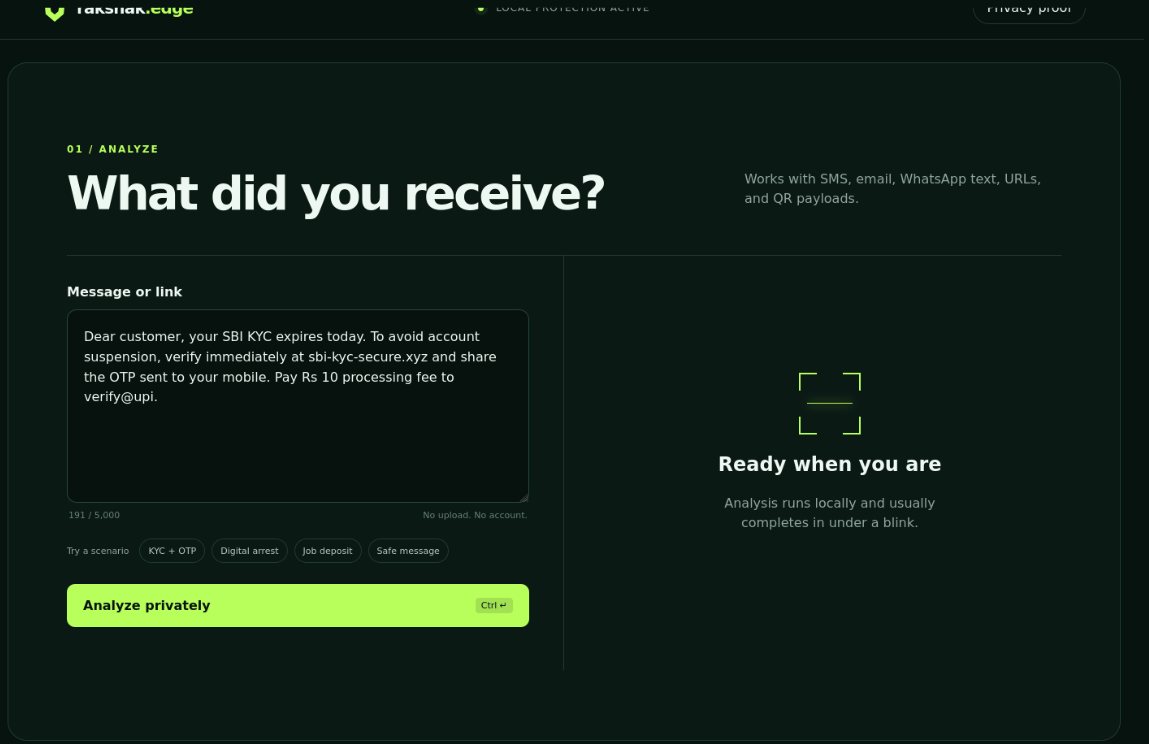
Opaque verdicts

A score alone cannot teach the user what triggered the warning or what to do next.

Design question

Can a PC explain the risk locally, in the moment that matters?

One local workflow from suspicion to safe action



Paste

Message, URL, UPI request, or QR payload

Understand

Risk score with visible evidence and extracted entities

Act safely

Independent verification path without another risky link

Hybrid reasoning keeps security explainable

Transparent signals stay authoritative. A compact classifier adds semantic context and runs on the Hexagon NPU.

01

Input boundary

Loopback API
No persistence

>

02

Evidence engines

URL + entity rules
Local semantic model

>

03

Snapdragon profile

INT8 encoder
ONNX Runtime + QNN

>

04

Decision layer

Calibrated score
Reasons + safe action

Graceful fallback

QNN unavailable? ONNX Runtime uses CPU. Model absent? The deterministic engine still delivers a complete, testable product.

AI catches the scam without a malicious link

72 /100

HIGH RISK

Stop and verify independently

Semantic scam pattern	99.8% high-risk confidence	AI
Voucher purchase	Financial coercion	RULE
No URL	Nothing for a link scanner	0
No OTP or UPI	Meaning comes from context	0

SAFEST NEXT STEP

Call the finance head using the number already in your contacts.

Measured locally, designed for Snapdragon

94.44%

fixed holdout accuracy

18

unseen holdout messages

7

regression tests passing

45 TOPS

Snapdragon X Elite NPU

Proof today and an NPU upgrade path

The repository includes the train and holdout corpus, the evaluation script, and the single disclosed model error. The Snapdragon profile upgrades the compact reference model to an INT8 encoder while keeping the same local API and evidence layer.

Holdout: high 6/6, low 6/6, medium 5/6. No cloud call during inference.

Working product with measured evidence

01	**Runs today**	One command, Python standard library, offline demo
02	**Measured**	Transparent 72-train / 18-holdout corpus and disclosed error
03	**Accessible**	Keyboard flow, high contrast, reduced motion, plain language
04	**Deployable**	QNN adapter, INT8 model contract, target acceptance checks
05	**Responsible**	Threat model, bounded claims, no server-side history

```
python app.py → localhost:8765
```

RAKSHAK EDGE

A safer pause before the tap

Private by design. Explainable by default. Ready for Snapdragon.

LIVE DEMO

```
python app.py
```

GitHub-ready repository | Pitch deck | Brief | Demo script

Hariharan C